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\DS\ASSI-2\A2-1.C 1

```c
//name     : Nirupa kutana
//roll no  : 2123
//div      : B
//assignment 2 :- 1
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *head,*newnode;
void create()
{
    struct node *temp;
    int choice=1;
    while(choice)
    {
        newnode=(struct node*)malloc(sizeof(struct node));
```

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F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

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\DS\ASSI-2\A2-1.C 1=[↕]

```c
int choice=1;
while(choice)
{
    newmode=(struct node*)malloc(sizeof(struct node));

    printf("\nenter data : ");
    scanf("%d",&newmode->data);
    newmode->next=0;

    if(head==0)
    {
        head=temp=newmode;
    }
    else
    {
        temp->next=newmode;
        temp=newmode;
    }
    printf("\ndo you want to continue(1/0) : ");
    scanf("%d",&choice);
}
```

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F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

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[■] \DS\ASSI-2\A2-1.C 1=[↕]

```c
        printf("\ndo you want to continue(1/0) : ");
        scanf("%d",&choice);
    }
}
void display()
{
    struct node *temp;
    temp=head;
    while(temp!=0)
    {
        printf("\n%d",temp->data);
        temp=temp->next;
    }
}
void main()
{
    clrscr();
    create();
    display();
    getch();
}
```

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F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

```
enter data : 10

do you want to continue(1/0) : 1

enter data : 10

do you want to continue(1/0) : 1

enter data : 20

do you want to continue(1/0) : 0

10
10
20_
```

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\DS\ASSI-2\A2-2.C 2=[↕]

```c
//Name : Nirupa kutana
//Rol no. : 2123
//Div  : B
//assignment 2 :- 2
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *head,*temp,*newnode;
void bins();
void lins();
void display();
void main()
{
        int choice=0;
        clrscr();
        while(1)
```

1:1

F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

```
{
        printf("\n1.first insert");
        printf("\n2.last insert");
        printf("\n3.display");
        printf("\n4.exit");
        printf("\nenter choice : ");
        scanf("%d",&choice);
        switch(choice)
        {
                case 1:
                bins();
                break;

                case 2:
                lins();
                break;

                case 3:
                display();
                break;
```

```
                        case 4:
                        exit(0);
                        break;


                }
        }
}
void bins()
{
    struct node *newnode;
    int item;
    newnode=(struct node*)malloc(sizeof(struct node));

    printf("enter value : ");
    scanf("%d",&item);
    newnode->data=item;
    newnode->next=head;
    head=newnode;
    printf("\nnode inserted");
}
```

```c
void lins()
{
    int item;
    newmode=(struct node*)malloc(sizeof(struct node));

    printf("\nenter value : ");
    scanf("%d",&item);
    newmode->data=item;
    if(head==NULL)
    {
        newmode->next=NULL;
        head=newmode;
        printf("\nnode inserted");
    }
    else
    {
        temp=head;
        while(temp->next!=NULL)
        {
            temp=temp->next;
        }
```

```
        temp=head;
        while(temp->next!=NULL)
        {
            temp=temp->next;
        }
        temp->next=newnode;
        newnode->next=NULL;
        printf("\nnode inserted");
    }
}

void display()
{
    struct node *temp;
    temp=head;
    while(temp!=NULL)
    {
        printf("\n%d",temp->data);
        temp=temp->next;
    }
}
```

```
1.first insert
2.last insert
3.display
4.exit
enter choice : 1
enter value : 10

node inserted
1.first insert
2.last insert
3.display
4.exit
enter choice : 2

enter value : 20

node inserted
1.first insert
2.last insert
3.display
4.exit
enter choice : _
```

```
node inserted
1.first insert
2.last insert
3.display
4.exit
enter choice : 2

enter value : 20

node inserted
1.first insert
2.last insert
3.display
4.exit
enter choice : 3

10
20
1.first insert
2.last insert
3.display
4.exit
```

\DS\ASSI-2\A2-3.C

```c
//name     : Nirupa kutana
//roll no  : 2123
//div      : B
//assignment 2 :- 3
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *head,*temp,*newnode;

void create()
{
    struct node *temp;
    int choice=1;
    while(choice)
    {
        newnode=(struct node*)malloc(sizeof(struct node));
```

1:1

```
        printf("\nenter data : ");
        scanf("%d",&newmode->data);
        newmode->next=0;

        if(head==0)
        {
            head=temp=newmode;
        }
        else
        {
            temp->next=newmode;
            temp=newmode;
        }
        printf("\ndo you want to continue(1/0) : ");
        scanf("%d",&choice);
    }
}


int count()
{
```

[■] \DS\ASSI-2\A2-3.C 2=[↕]

```c
int count()
{
    int count=0;
    struct node *temp;
    temp=head;
    while(temp!=NULL)
    {
        count++;
        temp=temp->next;
    }
    return count;
}


void rins()
{
    int i=1,loc,item;
    struct node *newnode,*temp;
    newnode=(struct node*)malloc(sizeof(struct node));

    if(head==NULL)
```

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```
if(head==NULL)
{
    printf("overflow");
}
printf("enter location after which you want to insert : ");
scanf("%d",&loc);
if(loc>count())
{
        printf("invalid location...");
}
else
{
    printf("\nenter value : ");
    scanf("%d",&item);
    newnode->data=item;
    temp=head;
    while(i<loc)
    {
        temp=temp->next;
        i++;
    }
```

```
        newnode->next=temp->next;
        temp->next=newnode;
    }
}

void display()
{
    struct node *temp;
    temp=head;
    while(temp!=NULL)
    {
        printf("\n%d",temp->data);
        temp=temp->next;
    }
}

void main()
{
        clrscr();
        create();
        rins();
```

```
}

void display()
{
    struct node *temp;
    temp=head;
    while(temp!=NULL)
    {
        printf("\n%d",temp->data);
        temp=temp->next;
    }
}

void main()
{
        clrscr();
        create();
        rins();
        display();
        getch();
}
```

```
enter data : 10

do you want to continue(1/0) : 1

enter data : 20

do you want to continue(1/0) : 1

enter data : 30

do you want to continue(1/0) : 0
enter location after which you want to insert : 2

enter value : 40

10
20
40
30
```

[■] \DS\ASSI-2\A2_4.C 3=[↕]

```c
//Name : Nirupa kutana
//Rol no. : 2123
//Div  : B
//assignment 2 :- 4
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
struct node
{
        int data;
        struct node *next;
};
struct node *head,*temp,*newnode;
void bins();
void bdel();
void edel();
void display();
void main()
{
        int choice=0;
        clrscr();
```

1:1

```
while(1)
{
        printf("\n1.insert");
        printf("\n2.beginning delete");
        printf("\n3.last delete");
        printf("\n4.display");
        printf("\n5.exit");

        printf("\nenter choice : ");
        scanf("%d",&choice);
        switch(choice)
        {

                case 1:
                bins();
                break;

                case 2:
                bdel();
                break;
```

```
                    case 3:
                    edel();
                    break;

                    case 4:
                    display();
                    break;

                    case 5:
                    exit(0);
                    break;
                }
        }
}

void bins()
{
    struct node *newmode;
    int item;
    newmode=(struct node*)malloc(sizeof(struct node));
```

[■] \DS\ASSI-2\A2_4.C 3=[↕]

```c
void bins()
{
    struct node *newmode;
    int item;
    newmode=(struct node*)malloc(sizeof(struct node));

    printf("enter value : ");
    scanf("%d",&item);
    newmode->data=item;
    newmode->next=head;
    head=newmode;
    printf("\nnode inserted");
}


void bdel()
{
    struct node *temp;
    if(head==NULL)
    {
        printf("\nlist is empty");
```

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```
    else
    {
        temp=head;
        head=temp->next;
        free(temp);
        printf("\nnode deleted from the begining...");
    }
}


void edel()
{
    struct node *temp,*prev;
    if(head==NULL)
    {
        printf("list id empty");
    }
    else if(head->next==NULL)
    {
        head=NULL;
        free(head);
```

```
        printf("\ndeleted node from last....");
    }
    else
    {
        temp=head;
        while(temp->next!=NULL)
        {
            prev=temp;
            temp=temp->next;
        }
        prev->next=NULL;
        free(temp);
        printf("\ndeleted node from last...");
    }
}

void display()
{
    struct node *temp;
    temp=head;
    while(temp!=NULL)
```

```
        while(temp->next!=NULL)
        {
            prev=temp;
            temp=temp->next;
        }
        prev->next=NULL;
        free(temp);
        printf("\ndeleted node from last...");
    }
}

void display()
{
    struct node *temp;
    temp=head;
    while(temp!=NULL)
    {
        printf("\n%d",temp->data);
        temp=temp->next;
    }
}
```

```
1.insert
2.beginning delete
3.last delete
4.display
5.exit
enter choice : 1
enter value : 10

node inserted
1.insert
2.beginning delete
3.last delete
4.display
5.exit
enter choice : 1
enter value : 20

node inserted
1.insert
2.beginning delete
3.last delete
4.display
5.exit
```

```
node inserted
1.insert
2.beginning delete
3.last delete
4.display
5.exit
enter choice : 2

node deleted from the begining...
1.insert
2.beginning delete
3.last delete
4.display
5.exit
enter choice : 4

10
1.insert
2.beginning delete
3.last delete
4.display
5.exit
```

```
node inserted
1.insert
2.beginning delete
3.last delete
4.display
5.exit
enter choice : 3

deleted node from last...
1.insert
2.beginning delete
3.last delete
4.display
5.exit
enter choice : 4

30
1.insert
2.beginning delete
3.last delete
4.display
5.exit
```

\DS\ASSI-2\A2_5.C

```c
//Name : Nirupa kutana
//Rol no. : 2123
//Div  : B
//assignment 2 :- 5
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *head,*temp,*newnode;

void create()
{
    struct node *temp;
    int choice=1;
    while(choice)
    {
        newnode=(struct node*)malloc(sizeof(struct node));
```

1:1

```
        printf("\nenter data : ");
        scanf("%d",&newmode->data);
        newmode->next=0;

        if(head==0)
        {
            head=temp=newmode;
        }
        else
        {
            temp->next=newmode;
            temp=newmode;
        }
        printf("\ndo you want to continue(1/0) : ");
        scanf("%d",&choice);
    }
}


int count()
{
```

\DS\ASSI-2\A2_5.C

```c
int count()
{
    int count=0;
    struct node *temp;
    temp=head;
    while(temp!=NULL)
    {
        count++;
        temp=temp->next;
    }
    return count;
}

void rdel()
{
    struct node *temp,*nextnode;
    int loc,i=1;
    temp=head;
    printf("\nenter the location of the node which you want to delete : ");
    scanf("%d",&loc);
    if(loc>count())
```

62:1

```
    if(loc>count())
    {
        printf("\ninvalid location...");
    }
    else
    {
        while(i<loc-1)
        {
            temp=temp->next;
            i++;
        }
        nextnode=temp->next;
        temp->next=nextnode->next;

        free(nextnode);
    }
}



void display()
```

```c
void display()
{
    struct node *temp;
    temp=head;
    while(temp!=NULL)
    {
        printf("\n%d",temp->data);
        temp=temp->next;
    }
}

void main()
{
        clrscr();
        create();
        rdel();
        display();
        getch();
}
```

```
enter data : 10

do you want to continue(1/0) : 1

enter data : 20

do you want to continue(1/0) : 1

enter data : 30

do you want to continue(1/0) : 0

enter the location of the node which you want to delete : 2

10
30
```